Elec-S1 Cheat Sheet – Sections 1A to 2C

Voltage · current · circuit vocabulary · power · characteristics · Ohm · Joule · Pouillet

1A — Electric voltage & sources

Box 1 – Voltage. Only potential *differences* can be measured (like height differences in geography):

$$u_{AB} = V_A - V_B \qquad u_{AB} = -u_{BA}$$

- Signed quantity: $u_{AB} > 0 \iff V_A > V_B$.
- Unit: **volt** (V). Potential and voltage have the same dimension.
- **Ground** = the reference point, where we set $V = 0$.
- Arrow convention: the arrow for u_{AB} points **towards A** (head at A, tail at B).

Orders of magnitude: neuron $\sim 100\,\text{mV}$; AA cell $1.5\,\text{V}$; Li-ion 3.6–$4.2\,\text{V}$; mains $230\,\text{V}$; HV line 25–$400\,\text{kV}$.

Box 2 – Ideal voltage source: maintains a fixed voltage e (the **e.m.f.**) across its terminals, *whatever the current* it delivers. Real sources have an internal resistance, so their voltage drops as i grows.

e.m.f. is in **volts**, it is **not** a force (no newtons!).

Voltmeter: connected **in parallel**; COM on the − side (else sign flips); ideal $R_V \to \infty$ (1–10 MΩ). Start on the largest range, then go down.

1B — From charge to electric current

Materials: conductors (Cu, Al, Ag, graphite: many mobile carriers), insulators (glass, plastic, air), semiconductors (Si, Ge) in between.

Origin: closing the circuit confines the field inside the wire; electrons acquire a **drift velocity** $\vec{v}$ *opposite* to $\vec{E}$. With no field, motion is random (thermal), net velocity $= 0$.

Box 3 – Electric current: charge dq crossing a cross-section S during dt:

$$i = \frac{dq}{dt} \qquad \text{unit: } \textbf{ampere} \text{ (A)},\ 1\,\text{A} = 1\,\text{C}\,\text{s}^{-1}$$

- Same current all along a series path (flow rate of charge).
- **Conventional direction** = that of *positive* charges ⇒ opposite to electron motion in a metal.

Box 4 – Microscopic expression. Carriers in the slice $S\,dx$ cross S in dt (with $dx = v\,dt$):
$dq = n\,q\,(S\,v\,dt) \Rightarrow$

$$i = n\,q\,v\,S \qquad (\text{electrons: } |i| = n\,e\,v\,S)$$

n = carrier density (m^{-3}), q = carrier charge (C), v = drift velocity (m/s), S in m^2. $e = 1.6 \times 10^{-19}\,\text{C}$.

Worked example (lecture): Cu, $S = 1\,\text{mm}^2$, $i = 1\,\text{A}$, $n = 8.5 \times 10^{28}\,\text{m}^{-3}$:

$$v = \frac{i}{neS} = \frac{1}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}}$$

$$v \approx 7 \times 10^{-5}\,\text{m/s}$$

Drift velocity $\sim 0.1\,\text{mm/s}$ is **not** the signal speed: the *field* sets up almost instantly, so all electrons start moving together.

Ammeter: connected **in series** (open the branch); ideal $R_A \to 0$.

1C — Circuits & two-terminal components

Box 5 – Vocabulary:

- **Node:** junction of **at least 3** conductors.
- **Branch:** portion of circuit between two (consecutive) nodes; one current.
- **Loop:** any closed path. **Mesh:** elementary loop containing no other loop.

Counting method (as graded on Test 1):

1. Merge every point joined by plain wire into **one** electrical node.
2. Keep only junctions where $\geq$ 3 components/branches meet ⇒ nodes.
3. A junction of just 2 components is **not** a node: they are in series, same branch.
4. Check: $\sum$(branches at each node) $= 2 \times$ (number of branches).

Test 1 circuit (E + 3 lamps): **2 nodes, 4 branches**.

Active components supply energy (sources); **passive** ones receive/store/dissipate it (R, C, L, lamp).

Box 6 – Combinations:

- **Series:** *same current*; voltages add: $u = u_1 + u_2$ (loop/mesh law).
- **Parallel:** *same voltage*; currents add: $i = i_1 + i_2$ (node law).

Direction: outside the source, conventional current flows from ⊕ to ⊖ (electrons go the other way, since $i = -nevS$).

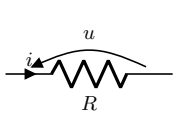
Box 7 – Electrical power:

$$p = u\,i \qquad 1\,\text{W} = 1\,\text{V} \times 1\,\text{A}$$

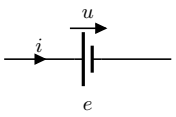
$$E = \int_{t_1}^{t_2} u\,i\,dt \xrightarrow{\text{DC}} E = u_0\,i_0\,\Delta t \quad \text{(J)}$$

Receiver convention: $p > 0$ = power *received*. Generator convention: $p > 0$ = power *supplied*.

Receiver vs generator convention



Receiver
u, i arrows *opposite*
current *enters* by +
$u = Ri$, $p = ui$ **received**



Generator
u, i arrows *same way*
current *leaves* by +
$p = ui$ **supplied**

2A — Characteristic curve & evidence for Ohm's law

Box 8 – Characteristic: the curve $i = f(u)$ (or $u = g(i)$) obtained by measuring pairs (u_k, i_k) for many operating states. It **depends on the chosen convention**.

- Passive component → receiver convention. Active → generator convention.
- Set-up: vary the source, read u_k (voltmeter, parallel) and i_k (ammeter, series); *short-shunt* arrangement.

Box 9 – Resistor: characteristic is a **straight line through the origin**. This is the signature of an ohmic conductor.

Slope of $i = f(u)$ is $G = 1/R$
Slope of $u = g(i)$ is R

Non-ohmic examples: ideal source (vertical/horizontal line), lamp (curved), diode, Zener diode, ideal current source.

Ohmmeter (Ω in a circle): power **off**, supply **disconnected**, at least one terminal of R **disconnected**, right range.

2B — Ohm's law, conductance, Joule's law

Box 12 – Ohm's law (ohmic conductor, **receiver** convention):

$$u = R\,i \qquad R \text{ in ohms } (\Omega)$$

$$\text{Conductance: } G = \frac{1}{R} \quad \text{in } \textbf{siemens } (\mathrm{S}) \qquad i = G\,u$$

- In **generator** convention the sign flips: $u = -R\,i$.
- An **ohmic conductor** = a component whose law is $u = Ri$ (linear, through the origin). Every real component (even a wire) has some R.
- Resistor: sets the current for a given voltage (values read from colour codes). Thermistors and photoresistors are ohmic too.

Typical R: lab lead $\sim$ tens of mΩ; lab resistor 1 Ω–1 MΩ; dry skin $\sim$ kΩ; voltmeter input 1–10 MΩ.

Joule effect: current through R converts electrical energy into heat.

Box 13 – Derivation of Joule's law:

1. Power received (receiver convention): $p = u\,i$.
2. Ohm: $u = R\,i \Rightarrow p = (Ri)\,i = R\,i^2$.
3. Or $i = u/R \Rightarrow p = u \cdot \dfrac{u}{R} = \dfrac{u^2}{R}$.

$$p = u\,i = R\,i^2 = \frac{u^2}{R} \qquad E_{heat} = R\,i^2\,\Delta t$$

It is $R\,i^2$ **never** $i\,R^2$ (units: $\Omega \cdot \mathrm{A}^2 = \mathrm{V} \cdot \mathrm{A} = \mathrm{W}$ ✓).

Uses: heaters, hotplates, fuses. Downside: losses in power lines and electronics.

2C — Pouillet's law, resistivity, conductivity

Box 14 – Pouillet's law (uniform wire of length L, cross-section S):

$$R = \rho\,\frac{L}{S} \qquad \rho\text{: resistivity in } \Omega \cdot \mathrm{m}$$

Longer wire $\Rightarrow$ larger R. Thicker wire $\Rightarrow$ smaller R. Circular wire: $S = \pi d^2/4$, so doubling d divides R by 4. ρ is measured with the *four-wire method*.

Box 15 – Conductivity:

$$\sigma = \frac{1}{\rho} \qquad \text{unit: } \mathrm{S\,m^{-1}} \qquad \left(R = \frac{L}{\sigma S}\right)$$

Good conductor $\iff$ small ρ $\iff$ large σ.

Material	ρ (Ω m)	σ (S m^{-1})
Silver	1.6×10^{-8}	6.3×10^{7}
Copper	1.7×10^{-8}	5.9×10^{7}
Aluminium	2.8×10^{-8}	3.6×10^{7}
Iron	1.0×10^{-7}	1.0×10^{7}
Graphite	$\sim 3 \times 10^{-5}$	$\sim 3 \times 10^{4}$

For most metals, ρ **increases with temperature** (hot wire conducts slightly worse).

Units, conversions & exam traps

G $= 10^9$ M $= 10^6$ k $= 10^3$ m $= 10^{-3}$
$\mu = 10^{-6}$ n $= 10^{-9}$ p $= 10^{-12}$

$1\,\mathrm{mm^2} = (10^{-3}\,\mathrm{m})^2 = \mathbf{10^{-6}}\,\mathrm{m^2}$ $\quad 1\,\mathrm{cm^2} = 10^{-4}\,\mathrm{m^2}$
To convert *to* a smaller unit, multiply: $3.2 \times 10^{-8}\,\Omega\,\mathrm{m} = 32\,\mathrm{n}\Omega\,\mathrm{m}$.

- Convert mm^2 $\rightarrow$ m^2 **before** using $R = \rho L/S$ or $i = nevS$.
- Node = ≥ 3 components meeting; 2 components at a junction = same branch.
- $u = Ri$ only in receiver convention; state the convention.
- Always give the **unit** and a sensible number of sig. figs.
- Definitions: give *words + equation + SI unit* (that's how marks are split).